

Bridging Brain Health and Mental Health - Georg Northoff

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foreign [Music] hungry foreign America it's your time I continuous make jokes sorry perfect do you want to sure yes thank you very much for introducing also thank you very much for inviting I had a lot of very exciting meetings so thank you for that I really appreciate it on Friday and yesterday so and one notable thing is really that you know what Dynamics is very simple because in many parts of the world people don't really know what Dynamics is um so we have a broad research program which really goes about brain mind connection and I show you today one particular line of that and that concerns psychiatry so we have this and I will explain this is um model of brain mind connection um that's a special Temple connection and from there we basically uh investigate also different psychiatric phenomena and the main idea here is for psychology is really that we can say this goes into the sum of the tentatively the mechanism of course I look at the computational people and they say this is not mechanism what you're talking about and they write probably least tentative and second that that also carries clinical significance meaning diagnostic markers and therapy suggestions to which I come at the end and I think that's why we really convert that you pursue a similar line of research so fire okay so thank you very much I didn't know that you do the job then you want to talk for me okay good so the monkey is and that's a little bit conceptual background so that you see where I'm coming from because that's very important to understand or not misunderstand what I'm saying so here uh what we often do in current Neuroscience uh it's really we are on the surface of the iceberg we look of certain cognitive functions attention working memory perception motor action which are observable to the outside and then we look for new correlates and then firstly that's the brilliant and said okay there's a computational mechanism predictive coding active infant generative model which I would say operate here and then you say no no no no this is not really true then you would say uh Frozen comes with the argument of theology but you would say the free energy should be between here and here

where it directly interfaces to the water but that's not true the free energy is calculated at the sum total of the prediction error so it's ultimately here so you need to really go into some of the operational details so I'm looking here why because I think we need to go back to the deeper layers of the brain where it directly interface or aligns you will hear that often or in quotes it's direct environmental input so again EAC encoding here is a decoding of what is encoded yeah and I'm looking here and you will see this there's a constant scene of my uh talk and it's important for Psychiatry and also I think it provides new views on the modeling so you will see we do a lot of input modeling so and what characterizes these deeper layers where I say these are key for the mind for the brain mind connection so that's really a different approach when you look in the current Neuroscience in the history of psychology and you'll say they always look for the mind up here I can't can't even reach it yeah uh they always look here I look here so I go backwards rather than forwards next one so that's basically an overview um so I first before going into the crisis of Psychiatry uh that I go into the special Temple approach what it is then I'll show you two examples of the spatial temporal approach depression with the speed disturbance I know this is all very strange for you if you're coming from cognitive psychology or cognitive behavior therapy or from biological Psychiatry what is speed it's trivial runs schizophrenia as a temple Precision disturbance and you will see I try to assemble a lot of evidence for that and then I come to the conclusions from mechanism to diagnosis and therapy next one so a very short brief in Psychology we even face more of a crisis than a neurology and where you're dealing with uh because next one um so normally don't know this a quarter of our mental disorder of all disorders are mental disorders yeah I mean sorry to say the numbers for dementia are low I think it's 20 30 million compared to depression anxiety is 80 milligrams look at the funding just as a side note yeah the discrepancy so we have to change that because uh the incidence of mental health disorders is increasing in particular very alarming among young people in particular after the coffee so this is not so in 10 years there was an estimate from the who it will be by 30 of all physical diseases not just 25 and when you ask around it's amazing I remember that last year in October there was an article in the in the in the Canadian newspaper about our work we were on front page and in the evening I went to the supermarket and brought the newspaper and then the cashier looked at me no this is you I said she's asking what are you doing yes I'm dealing with anxiety and breathing service oh I need that too yeah so even the cashier but just by chance is unbelievable yeah uh and eighty percent of many of them are anxiety and depression yeah so it's it's uh really tendency increasing so we focus our main focus here for the sake of

pragmatism but it will be extended in the future is anxiety disorders depression uh schizophrenia bipolar disorder but we'll also include for instance post-traumatic stress disorder because it's very important next one so uh what is the challenge of treating mental health disorder we have no markers we don't even have neurological Sciences you have you have some symptoms which uh many psychiatrists interpret in different ways so there's a saying if you attend psychiatrist and one patient you have 20 diagnosis yeah so it's really it's it's very problematic there's a lot of discussion about classification DSM some of you might heard that the alternative systems yeah so that's on the diagnostic side so we have more diagnostic markers yeah so it's a catastrophe then depending upon your orientation where you're psychodynamic or whether you're Freudian or Union or whether you're biological you're diagnosed different the catastrophe and that continues with the drugs I'm not against drug treatment so if there's some pharmaceutical company people here fair but the drugs are they suppress the whole brain they're not selective so imagine if you have a broken bone you're in your lower right leg and you need to do some physiotherapy in order to keep the rest the the upper side and the other leg intact now the doctor puts a cast on both legs up to your body so that's basically what we as psychiatrists do yeah it was many drugs so and we have a lot of trial and other treatments and of course I don't need to lead to a lot of suffering next one so why is that because we don't know the mechanism we have no idea about how this we see abnormal neuronal activity on a group level many results cannot be replicated let alone individual prediction you have data sets with thousand ten thousand of subjects let's say in imaging or genetic in media music so the question is whether you just increase your case numbers and use more machine learning methods or whether you think about your model of your disease on which you base your investigations so you will see that I go the latter way yeah and I don't need to say the other way is often technology latest is now computational Psychiatry we get into mechanism but it's the same old model yeah everyone they have the same crisis they don't yield markers um so this is clear unlike in Physical Medicine maybe even worse than you dementia guys next one so here we summarize the distance from a recent paper which will come out in schizophrenia bulletin so it's a twofold it's a crisis of mechanism we don't really understand the neuronal and computational mechanism of the psychopathological certainly yes we have the predictive coding we have empirical prior but these are unspecific changes across all disorders empirical prior is abnormal or not yeah so it's a real lack of specificity as brilliant as resistance model is and I would really say it's a brilliant thing so we have no biomarkers here on the left I'm here um that's also a stagnation in the

development of new psychotropic drugs and other therapies yeah so now many psychiatrists just try out some therapy yeah because that's it that you you get grants but again it's a blind so some patients will benefit some not and the results are usually immediate forget it and then that's a specific twist I would like to add as I'm originally from Germany Continental European it's a crisis of subjectivity because what these psychiatric disorders are essentially subjective disorders it's your experience so I can hear voices but okay I can function very well that the way I experience them subjectively that's the keys yeah so auditory hallucination is a completely unspecific symptom you have it in dementia often in Progressive stages of dementia you have auditory hallucination yeah and here here there we are so same thing here so that's why we really need to move the mechanism with the subjectivity and that's the background question for the approach I'm doing next one so what we link is this what we miss is this triangle between symptoms brain and experience and that's what I call among others common currency that's the key that's basically now if you're not interested in the solution you can go home now and just say that's what I'm talking about next one okay I say next one so now I show you a little bit what's the spatial Temple approaches so it has his roots also in philosophy because I'm also philosopher I don't want to go into philosophy now but I want to tell you one hidden preposition in current neuroscience and psychiatry that's the present position of what I call the presupposition of specialness so the mind and mental features and Consciousness and self are something special there's a special mechanism in the brain added on to the other non-special mechanisms so here in philosophy you have is this the red stuff it's reflective and phenomenal Consciousness and I block access to normal consciousness I explicated that for the Consciousness Theory but it goes the same for the self and all other mental features and we see here in the Consciousness Theory uh integrated information it's a special mechanism slash integration on top of the non-integration in the brain that's completely stupid sorry I hope there's no IIT proponent here yeah okay there's a special embodiment and then we have Consciousness as if the brain is not always embodied only the upper part of the iceberg has information integration only the other part is yeah uh enter a higher order sort Theory now Rosenthal uh Global neural workspace and this is a recurrent processing Theory by um yeah and then of course here is the brain and that person really says the only model of brain is basically really the friction Theory and I think also Marcus rack is somewhat but it's not well elaborated his interesting model yeah and so this is and that's a month but I don't need to tell you is for those who are Western philosophy that's basically a continuation of streetcar with the Neuroscience it's a Cartesian Neuroscience it's a dualism yeah so there is something

special to the mind and let's place it yeah okay now the people say no no no no not after you cannot say this but you will assume something special in The Breakfast itself that's neuron Cartesian next one yeah and that leads to all the problems discussed in current philosophy the icons is the gap of contingency uh it's the explanatory Gap problem it's a hard problem David Chalmers and so on and so forth yeah Big Splash no output sorry uh I will be fast so guys good so next one I told you okay now so what I do I say okay maybe we should not look for the difference we look for something similar what is shared again and this is it it sounds strange and it sounds very metaphysical no it's basically what when you look into the history of science that's what people do they look for something similar David looked for what different speeches share what's a human shared with the monsoon thank you um so what humans and non-humans share I mean from a Christian perspective was a catastrophe yeah uh looked for what the Earth share with the other planets in the universe not what makes it special what makes it different so this is really and you see in some of my books I claim for a companion term but this so I look for the similarity so now what is similar so I always like this quote uh by Nikola Tesla so it was an engineer uh 19th Century or from former Yugoslavia and then he went to New York and look at his uh Google very interesting guy a very interesting very idiosyncratic and there's no I think there's no coincidence that uh Tesla uh that have a mask Elon Musk name discuss Tesla I think they share the Innovation the really creative capacity which is amazing but also the idiosyncrasies in the personalities so that is really it's amazing so what he said this beautiful thing if you want to find the secrets of the universe think in terms of energy frequency vibration the brain is part of the universe the mental features are part of the universe so this is the way you will see next one I can't deconstruct basically the brain so for me and it's always a pleasure when I say okay the mental thing is missing especially the sun Dynamic is some slow time scale longer time scale it always puts you see a smile on my face uh because it seems like you don't need to think about something yeah and that's what we called in this 2020 paper which has taken up a lot fortunately uh comma currents so there'll be explicate that concept and I say this is what I call spatial temporary so spatial temporary Neuroscience it's not just okay there are some topography and Dynamics in the brain no that topography and Dynamics shapes your cognition shapes your mental state that special temperament yeah that transition is an explicit let's say if you want to say it's a brain mind approach and you will see I chose this for psychotic disorders I indicate that for dreams and for mind wandering we do this for self uh we have the temple special Theory Of Consciousness so it's all analysis runs this morning I had a zoom with somebody in my group

for working normally there's even the temporal envelope time scales for working normally of course make sense because it's a temporary phenomena it's not a special cognitive representation next one so that's a shared special topography on Temple Dynamic and now you will see that's the way I deconstruct now mental disorders so I always can see uh people who discuss with me particularly the mind one as the meditation I deconstruct that in terms of spatial topography and Temple Dynamic I infer happily uh back and forth between phenomenal slash experiential and neuronal level yeah we had to see irreversibility so I look for dynamic phenomenon maybe there's something on the experiential side which is often neglected but then I can also ask my patients whether they experience it and what there we go next one so so that's basically here as I said to indicated that what we developed this Temple special Theory Of Consciousness and first is really meaning spontaneous and stimulus related activity because that distinction is completely artificial anyway the blowing if it were sitting here it would smile how stupid we are by using the brain um and second it's an explicit link between neuronal and phenomena yeah uh I will I will give you later some examples about meditation and dreams as examples of that yeah very short uh I already have that here next one next one there must be another one yeah so here okay I already give you the two examples of the template spaces Theory houses neuronal phenomenal linkage work so this is meditation as a published uh paper we published end of last year uh so basically what we did it was done here by the first also awesome Cooper and Bianca Ventura um we investigated uh naive versus professional meditators FMI and we looked for the whole written pattern and one particular pattern was a topographic pattern but here's a balance between the Fortnight Network and control executive Network usually they stand in an anti-correlation negative correlation uh so if the dln is high the c n is low and vice versa and so that's basically the pattern was found in naive meditators however now when you go into uh profession so that's basically in that organization goes basically the cortical midline structures DNM health Team often associated with self-specific self-related processing so it's an internally oriented cognition whereas the control executive network is an externally oriented cognition it's goal oriented so when I now look for you I recruit more my control executive Network whereas if I'm more sort of here in myself and thinking in me undering how could I make my talk better that people understand me then it's more internal oriented and cortical midline structures so in that balance sort of trips its neither so it's a beginning it's like this and in the intermediate stages like this and then the professional meditators there's no difference anymore it's neither a positive or negative relationship so and then also other data which I don't present here we also see that the

relationship between DNA and visual cortex is much more synchronized the whole brain is much more unified synchronized and less differentiated in its topography so what does this mean for input encoding your input is no longer differentially encoded in the different regions the input remains as it is from the environmental input and that's why you probably soon colonize with the environment and that's a basic experience of proficient meditation you finally in tune with the environment you hear this you have other models you can have the the server who is in synchrony in tune with the wave the musician when you tap the beat to the music your synchrony your brain synchronizes and even I like that particular you showed in recent paper said if you have a certain Temple structure in your task 20 to 30 seconds you have a peak in exactly that frequency range in your power Spectrum kill the broom and caught the task periodicity that's why I always wants a temple structure of tusks it is often neglected in completely neglected it's much more clever than we think so it's amazing so here in professional meditators so what we said okay now you have a dual organization internally externally oriented cognition now on the psychological level and that goes along with the Dual organization of your brain Dion and Versace now in meditation you lose that dual Topography of your brain and that's exactly what you experience as non-duality so that's a big thing that you don't make here audio Duality self other impinges upon your belief upon your cognition but in the non-dual state and it's very well described in the middle letter which particularly when you go to India they're crazy about non-duality and non-locality this is unbelievable uh so you really see that Duality is no longer there and that no longer shapes your thoughts your environment so you don't make that distinction yeah the basic subjectivity is preserved but to serve nonsense dichotomy is no longer there the basic subjectivity is preserved your point of view as I would say but I don't want to go into them so and also interestingly see you'll see an enlargement of your Dynamic repertoire your Dynamic Lincoln had some nice papers on that uh with the detained fluctuation I notice you see you're really see an enlargement of your Dynamic variability and why is that important the more variable your brain the better you encode you can encode the variable inputs from the environment and that's so better you can align to the stochastics of this movement I work here if you have a large dynamic repertoire you can follow me yeah I hope so uh yeah um and that's also you better align with the environment but you process your end you encode more inputs and you process that input in a less differentiate way so basically you less manipulate input that's very well reflected in many meditative experiences next one so that's an example where how the normal topography translates into mental topography then that's you will see I make a similar case now for mental

disorders so for me psychiatric disorders are specific space-time disturbances and as you know space and time have multiple dimensions so that's of course I can say now they're spatial temporary disturbance but that doesn't heal anything clinically I need to specify and that's what I will show you in the following so for me mental disorders as a general statement of space-time disturbance of the brain's inner time in space maximum I provide a little bit uh flesh to the poems I'll show you two examples and the first example is depression is a speed disturbance next one so as you all know depressing with melancholic the typical there are different forms of depression I don't want to get particularly I focus here on the form of depression where you sit in the corner you can't get up in the morning you're depressed you have rumination everything focus upon yourself negative mood you don't move your perceptionist and so on and so forth so everyone the working assumptions this is based on on many years of studies and phenomenology experience and back and forth and Neuroscience in our show it's really fast it's a speed disturbance you're too slow your brain is too slow and your psychological functions are also too slow and your experience also reflected slow now and that's a huge this is for me basically when I speak of a basic disturbance or generative disturbance and one basically like the relationship between glucose and insulin is the generative disturbance for the various diabetes symptoms throughout the body the null comparison I always like to make that comparison so now comparison in mental disorders only in depression you have effective symptoms you have cognitive symptoms you have sensory systems you have motor symptoms you have social symptoms you have mind wondering problems you have basically all cognitive functions are changed yeah and they're not as strongly change let's say like in Dementia or like in the frontal Dimension it's more sort of a graceful degradation there's some computational it's even cross back I think he used the term as a graceful degradation of a network yeah so so it's some sort of subtle not clear stands highly inter-individually variable symptom constellation so for me it's not a single symptom it's a co-worker we call this co-occurrence of symptoms and how is that possible so as I said that's in the same way as in diabetes you have a co-occurrence of symptoms here in the retina here in the Victoria the gangreneur your symptoms everywhere now imagine if you weren't known that this is related to insulin glucose it would say these are all different disorders here are different people present you would not combine them yeah but once you know the basic generative disturbance the glucose level modulated by the insulin yeah makes sense so glucose goes everywhere and so on and so forth and the individual difference failure so that's what I call so be careful when I speak of generative or basic disturbance I do not mean the cause the

causes of diabetes are the causes of the depletion of insulin there are multiple costs could be secondary or you eat too many sweets environmental impact yeah very interestingly I just want to note that I come back yeah and it come out to be by the heritable type 1 diabetes and so forth I don't know much about diabetes as you see okay next one I just want to make the analogy I'm not doing anything special or metaphysically and next okay so now I show you the slowness in different domains psychological and brain so here this is a line of research is visual perception so we do this motion suppression task and it's intrinsically a dynamic task because you have basically these little uh gratings it's called they move and you have to click the button when you can see in which direction it moves so the task itself is dynamic and what we modified this is a standard task which is very well known to a particularly activate V5 middle Temple right higher order visual cortex uh motion suppression area so it's very well known and you can calculate it through some complex Circle physics uh which is too high for me so you can basically and reconstructed that how that perceives this so this is this is a new Point what we did because from these basically your reaction time and other parameters and how fast this move you can reconstruct how much the gray level of these pixel changes yeah compared to the original and then you can basically reconstruct how they perceived the original and that's important so this is what I call subjective reconstruction because that's what we need in Psychiatry how do people subjectively perceive this thing your objective measures are not relevant because you want to see how that perceives this thing so and then we also included a fast and a slow version and what we could see that the depressed process perceives the fast version likes a slow version so now imagine you cannot make a difference between this and between this what I did first was the same as of course you become depressed and you become detached because you can't perceive half of the reality so now I'm switching your attention to this one yeah and where you can reconstruct how much as I said the grow level of the pixel changes relative to the origin and what you can see and you can calculate that and curve if you called chords is motion distance and you can see that the difference between the original uh which is sort of somewhat somewhat static uh is much larger in the uh healthy so they're perceive it much more dynamic than the uh MDD so it's more static and that's exactly when you ask the people that's what they tell you yeah so it's no longer they lose in their visual cortex and when we showed that this is related to decreased uh Power in the faster frequencies in the exactly that area in empty this is various papers we published you see this over the last yeah this is for instance here so here we did a separate uh experiments so here we did uh visual checkerboard stimulation in different frequencies in four different frequencies you can

see this here on the right we stimulated in four to eight Hertz 1.2 Hertz 0.4 and 0 Hertz yeah so to stick about same stimulus just different frequencies so if we probe for how the external dynamics of the input modulates activity in particular here uh here in the empty area and this was fmri and you can see that in the healthy subjects uh you have more higher amplitudes of this glm modeling in FMI higher amplitude in the two faster frequencies no surprise because they recruit more neurons more activity so you have higher amplitude the blue ones are the depression and you can clearly see this here the depression are less and then we grooved the two slogans and the two fast run and you can see a clear difference the depressed do not process the fast ones in the empty it's more similar to the slow ones so one interestingly because this is really a task it has been shown in monkeys and animals and and humans this is really empty but we also show that other regions so sorry other regions are strongly involved the anterior cortical midline regions which are known to be key in depression so they're also involved and we can see the same thing here there's no difference how they process the Fast and the slow stimuli even where anterior cortical Midland so that really suggests Maybe that's a whole brain thing and we have another study which uh where we showed different subjects are very short the same thing next one um then next one so that's a therapeutic implications because you can single out patients who have particular Stone visual perception and V5 the normal difference but we and marker and I want to use it as such so that's why we did one study oxygen petal cortex TMS which in the field of depression is unheard of because it's everything does a lot of prefrontal Cortex and maybe a little bit frontal and if you do something else trust me they kill you it's unbelievable because there's a whole Mafia and there's money involved and FDA approval private practices and blah blah blah user American way of doing things so and not very good rates yeah 5066 reformers then we did another study particularly for motor cortex people and again we showed very good efficacy so end of the story probably doesn't matter where you apply the TMS files because the brain acts as a global brain next one you don't tell that to the team Institute good so so that's basically your show in input regions is too slow on psychological and neuronal levels that's why I'm so in the modeling I really you see this in some of the model stuff in our recent paper for input encoding so I'm interested in simulating different kinds of input constellations and different kinds of State dependencies how they relate to different inputs because that's for Wiki for me these disorders are encoding yourself so now we did something really interesting at least from my point of view thoughts so what we did we try to get into the thought Dynamics so this is a line of research which we're doing that we say we calculate that thoughts in terms of time

scales and more will come out next one so here we did a study where you can see this here so subjects were in front of the computer and they were instructed to run the exact content change from internal to external sort so external sorts whenever there's a view environment internal thoughts about my own self and then they had to click the button um and here we did it queue based you can also do this non-q base uh and let uh 10 20 30 and all in between uh second where we gave cues how much it was internally or externally oriented and we also measured the degree of awareness how aware are this thoughts we did this in healthy uh depressed ndd major depressive disorder and depressed bipolar patient and the first thing um maybe I show you next one and after that next one next one okay so now we did and we reconstructed that in terms of the power spectrum because you have a Time series of change and you can end your awareness ratings so the awareness ratings are basically uh uh um yeah and here you can basically calculate and you can reconstruct that we didn't fit here and we construct that time series and you have to do this for quite some time so this was 15 minutes because you need a number of Trials obviously because you want to reconstruct this so here's the frequency and here is the power so it's the frequency of your thought change internal external Soul change and the power associated with that so you could see here it helps the subject they have all frequencies and a fair amount of power Now look for the ndd patients no power and not much change ask the patients that's exactly what they experience I'm drained off I have no energy and nothing changes my thought is the same ruminating thought circular thought you see I indicated we correlate that with a breathing as a subset of rumination scale and exactly High correlate now in the depressed bipolar we saw a similar pattern but not exactly the same now you can go back click this one so here for instance I show you one main result of that study that's a frequency of the sword was summed up slower in the MDD than the duty so we have there are similarities between ndd and depressed bipolar which of course makes sense but also some so another possible differential diagnostic test here which doesn't exist in the literature next one now you can go ahead ahead next one so that's about sort we're thinking there's one study of course now you say do this for emotions yeah so that's something of course we still need to do but there's one study in pns 2014 input at all from a Dutch group it's very interesting they showed indeed that the Dynamics before the onset of depression you have a slowing in your change in your emotions on the psychological level biblical dynamic system critical slowing period yeah and that's exactly what you observe and there's other paper since on that so I showed you some dynamic and speed only psychological and partial neurology level nine I'll show you a little bit about the topography so I'm a big fan of what is called the

global signal in FMI do I know guys usually often money will just regress it out however it has a physiological basis has been shown in Monkey studies combined electrophysiologically fmri Leopold shermanik did some fantastic papers there and it's basically says there's an infraslow fluctuations That's the electrophysiological basis and what are infraslow features that long cycle durations they extend over several regions so Global is to be understood in a relative sense yeah so we have big form of the global signal um I think there was a no no no no no you're too fast for me and I'm too slow for you okay there we are that's a discrepancy um so okay so what we did so now say Okay more this is a uh remember what I showed in the meditations these are Global topographic new organization disorders yeah your topography reorganizes your balances we had one paper in 2016 in pns where we showed that the balance between DM and intensive motor network is abnormal in bipolar and differential cell in Mania and depression yeah so here um let me show you uh Global signal pattern we replicated those findings and others too that have increased Global representation of brain activity in the cortical midline structures instead of forward mode Network yeah and what do I mean by that so look for climate change bad topic but unfortunately very relevant you have a global change in the ozone layer and that is differentially Manifest in different parts of the world yeah that's exactly same with the brain you have Global activity shared by all regions of the brain but uh manifest to different degrees in different regions of the brain that's the global signal topography calculated gscore so in the two and depressions I show significantly higher Global signal activity here in the midline regions and then investing the proof of principle when you apply the global signal regression then that difference appears so that really suggests that there's too much Global activity representation in this Midland region everything is focused on that and the sensory regions remember the visual cortex have decreased Global signal representation so basically then it was a very clever thing done by Andreas calabrini a former postdoc of mine who is now professor in Bergamo um so then we correlated the GS in the uh Department network was a non-defod and you see an abnormal correlation yeah that is uh when you apply uh the in the healthy subject you don't see that reading the fart mode and unimodal regions are more independent in the healthy subject but in depression your unilateral regions are completely enslaved by the default mode your perception is enslaved by your illuminations by your focus on yourself you're negative your guilt and everything that's exactly the clinical reality then this was a small example 50 subjects um we did machine learning we got an accuracy of 90 with 50 subjects it's nothing you have studies with thousands and hundreds that don't even reach coming close next one so clinically relevance next one okay

now that's another diagnostic tool which I would like to use it's a global representation so here I show your data from uh that was a paper by Young head I 2017 in schizophrenia you see uh abnormal conversion of your Global signal representation between uni and transmodal regions then we saw them bipolar increased Global signal representation for among others in the motor cortex in Mania that's very interesting because when you see a manic patient she or he walks around like crazy so we have increased Global activity in the motor coding that correlates with your psychomotor agitation symptom yeah in depression we got the hippocampus we can see that here and again makes sense increased normal ritual autobiographical memory retriever yeah next one yeah so I don't know just go back yeah and then the findings I showed you before so you have three different patterns of the global signal topography in schizophrenia in bipolar in depression clinical biomarker next one so that's why we like to validate this now in large-scale clinical studies so same I showed you thoughts I showed you visual perception I showed you brain now I'll show you movements next one um so here this is a paper just came out last uh very end of last year so too fast for me I'm depressed from your perspective right I didn't know that I looked like that um so here we did a visual emotional Paradigm it's a two by two uh fast there's a slow and neutral versus negative stimuli because we monitor interaction between speed of the stimulus and the emotion so that's what we did so in particular and we apply the global signal stuff here so we did not apply the usual um uh amplitude based event related blah blah blah approach but we've applied a global signal approach so we presented that in block a design four blocks of these four conditions a negative fast negative slow neutral slow and fast because for Global signal you need to analyze in Block terms so do you need a continuous data set and maybe as somebody some water that's in between I can drink a little bit sorry um and so on what we observed here is uh but in short uh Global signal representation here in the uh different regions and in depression during the task particularly negative fast they should decreased Global signal representation so they cannot activate the global signal particularly in negative fast similar and it was always a global activity component when we did the GSR Global signal regression the findings disappeared as in the other paper so the test is important of the global activity component each region has two activity components a global and the local and that can shift and you see this typical regions posterior cingulate anterior midline regions here anterior midline posterior singular typical regions uh Superior Temple gyrus so like typical depression regions and then we did another analysis this is apostolio just whole brain coronation with psychomotor retardation because the idea was maybe that correlates with the

degree of psychomotor retardation in our scales and there we are we got again of course makes perfect sense we had the pre-central gyrus primary motor cortex inferior frontal gyrus posterior cingulate cortex correlating with motor retardation so now given these findings you would like to apply in subjects which show strong cycle to retardation and Global signal decreases in the motor cortex that's the people who got should get motor cortex DMS and we have one pilot study which is we're just preparing for submission where we successfully applied TMs in motor cortex of depression so here you have direct link of your symptom potential potential mechanism quote unquote plus a therapeutic indication next one so now you say okay uh how is all that related to your negative emotions so you want to know something wrong again here topography next one so and I have to make a little detour here um so the self is a big Topic in Neuroscience uh there are probably as many concepts of self as researchers around and it's a catastrophe so here we did the large scale motor analysis uh of all task evoked studies on self on the body on the interception and also on the typical ah perfect that is very nice thank you I appreciate it um let me just have that and so here we did a large scale meta-analysis or task evoked studies in a healthy subject and what we observed is sort of so we looked for various regions involved I don't go into details here about that um but we sort of observe based on the data three different layers so intersective regions typical insular I don't mention that here salamos as a subcortical regions because makes perfect sense that's all the interceptive tasks when we have interceptive awareness uh original heartbeat and all that when extraceptors this was the proprioceptive task mainly Ola blanca the studies an interesting resource studies found the temple periodical Junction you know all this premoto cortex anti-metal prefrontal cord but also replicated the same reason as in The Intercept so these were independent set of studies but these regions which you observed here were replicated here yeah plus additional reasons among the same these are the typical mental self-task like trade adjectives or autobiographical stories so it's about your cognitive or mental self again you saw to my delight of course also according to commit line structure plus the region from the previous level it depends that so that hints Upon A nested organization likes a Russian adult uh um the shape of the smaller one is replicated in the next larger one and so forth yeah and you have also in China you have beautiful Crystal Bots same thing these are really beautiful it's incredible yeah so you have a nested organization now what do you think this nested organization when some of this balance changes that's what happens in psychotic disorders next one now you're allowed to click okay perfect so now you need it so here we did again it's an FMI it's an FMI approach again this Global signal so we looked for the degree of global activity representation in stop this is tricky

I don't know why that is okay um some hidden artificial intelligence there um so so we basically investigate the degree of global signal representation in these different regions of the different layers of cells in healthy and in depression and this is FMI and what we observed and during the task during an emotional attention task and you could again see inter as okay X receptive and these were trait depressions these were non-acute MDD patients so this is trade features it's important it's non-acute and what you could see it's an increase in the X receptive and also an increase in dementia no surprise that goes well with the symptoms and here we uh it's the same for the different we plotted this is the same data plotted here but according just the sequence the balance between the three different layers and you see it's really very much different there's still these two layers Stand Out compared to the intercept so you see already that the growing signal represent in Roma I present overall Topography of DM and non-dmn now I present you within basically the transmodal regions I again show you an altered topography with an abnormal increase in the dmn and that's related to yourself stuff then we also did an emotional attention task um and the behavioral data of that I don't want to go into this this is the multinomial regression analysis so we had four values of the emotional attention task whether they could attend it or not positive negative and neutral and you could see that these different layers also modulated the emotional attention task and so meaning you have an abnormal Global signal representation across the three layers of cells and that in turn leads to other normal emotion regulation or dysregulation or emotional attention so here you see that the self might be on a psychological level disturbance because it's in Boston and everything so again these three layers that's what I want to develop it's probably hopefully a differential diagnostic marker for instance in anxiety disorders you see a decrease there okay okay so that's difficult then I speed up okay I finished the depression in time and then I um next one next one you already know this okay next one no go back so many more slides coming okay I hope you'll be taking a couple of more minutes okay good um so here you can say yes but maybe your time changes your speed change it's just a consequence of abnormal cognition that's the way cognitive people would argue for and I said no even on the psychological level your abnormal time in the brain constitutes or generates an abnormal time perception of slowness and that in turn shapes your cognitive function in an abnormal way and slash your symptoms so it's an abnormal Dynamic scaffold or envelope for cognitive functions so and yeah people would argue in a psychiatric process it's a Time perspective time perception is just a consequence of the depressive symptoms as I said I said the other way around so here we tested this pure psychological study 100 subject and I know I need to be

fast so we also investigated childhood trauma higher incidence in these depressed patients and we are two different mediation models easier time perspective zimbaro Intel perspective interval uh inventory either is a moderator or is a dependent variable so which model mediation model hold this one this yielded a full mediation this one not so if you have a trauma you might try or might not lead to depression of your time perspective your time perception is normal yeah and here this is the other way around this year the full mediation no mediation at all so this really support for me time perspective next one okay and we have another line of research again with my former post talking there will be a few different layers of trauma related to the three layers of self does this goes strong on the psychodynamics this is a collaborator here because she in from a purely psychodynamic perspective distinguish these three levels of trauma but I don't have time next one good I hope that's clear and I hope I could also indicate it has potential biomarker relevant next one so now this I will walk you very quickly through uh temple in Precision next one uh positive negative symptoms so the idea is here this is a temple in Precision you're imprecise next one I show you the data and I walk through so fast uh Timber windows are too long next so here we did also the next one uh we no no no stop uh we did also the uh autocorrelation window and we probe for temple integration if you integrate too much you're temporally imprecise yeah um so we prove for temporal integration on the marijuana level measuring the autocorrelation window uh and on the psychological level by this continuous task is a phase morphing task from the zakira's originally uh face more often task where you have to indicate when it's no longer your face or when it's your face so itself as a self morphing and in either direction you go and that's basically temporary integration because if you integrate too much you still see another person's face as your own face so that's a design because it's Temple integration which is in the Paradigm itself next one yeah that's the autocorrelation window uh basically you measure your time series with itself the decrease to which it correlates in terms of next next one uh yeah so here next one uh we saw that the autocorrelation window is too long in the schizophrenic patients and that is that was 2021 published you see this here significantly longer here the red ones compared to the blue ones and this is replicated in a completely independent data set which was just published a couple of months ago you again see the autocorrelation window doing the Oddball task and in the resting state too long so your two long Windows you integrate too much so we can temporally imprecise next one yeah I'm temporally imprecise now I know but I try to uh judge so how is that related to symptoms so here we were lucky we could collaborate here with Carlos and Denmark who developed this unfortunately David is

not here David Martinez I I'm sure he knows his scale for uh experience of their own self abnormal itself and that's a qualitative interview and what we could see that the relationship between the basic selfless order and schizophrenia and negative symptoms is mediated by your time scale so that is beautiful because you see during your own operates in the background and mediates the transition to the single that's exactly what I want that's why I speak of space or Temple psychotherapy next one okay next one I probably skipped yeah okay next one I next is very short so if you confuse internal and external reality then how's that related on the mironal side meaning you may not be able to make a proper distinction between your resting state or pre-stimulus activity and your task evoked activity moving your external task or input is not properly differentiated from your ongoing activity so my ex you you look at me but I just don't process that really properly just integrate it into my ongoing mentation my inner thought and of course then I think you will want to kill me or some strange stuff yeah so that's exactly so here we did the meta-analysis on different EEG that is mainly on also fmr and schizophrenia those studies which included both rest or pre-similar's activity and task revoked activity in the analysis and you will see it's not many studies this was truly 21. so catastrophe and reviewers ask me why are you doing this is not relevant it's incredible yeah so he showed this and whether the pre-stimulus measure is high or low his red stuff is too low or too high different kinds of EEG measures in each case you have decreased three stimulus task difference so that's really here system is an extreme Dynamic State too high or too low it doesn't react anymore and that's of course confusing the internal extern next one so this is a really interesting stuff you know I have to drink a little bit I hope that Melody will give me and then I continue here so here we look now for Temple and Precision in the millisecond range in the amplitude and in the face of the EEG signal so this here this was very dangerous was done by her from Greece I never can pronounce her name so it's Maria is her first name it's easier for me so and she applied his very clever approach uh it was an internal initiation Paradigm where you're based as you click and then you get a tone uh and she applied a temple transfer function is a very clever that and what you could see that could see in all these frequencies EEG obviously that you could see increased delays in the amplitude by applying this transfer function yeah and she also saw a increased intra-subject variability of those stimulus so that's the amplitude uh is uh modulated and more unstably delayed into schizophrenic and this is millisecond yeah so that's you could find out through the transfer function and again replicated so then we did for the face so in the amplitude you have some Temple and Precision in the millisecondary now interface this is intertrial Phase coherence

you have an auditory Oddball Paradigm continuous tones you have to click the button but you have to pick the button only when there is a deviant so it's called an attention Paradigm but for me it's an external synchronization pattern where you have to synchronize and you can see the phase onsets synchronize with the onset of the external stimulus the line so what you see in the triflex coherence in the healthy subject you see in particular inseta and Delta because that's the reason of the inter-trial interval of your tones nobody is mentioning that you see that a high phase coherence in the brain and you see the c as a typical phase Circle High now in schizophrenia you don't see that the phase can't synchronize and who sees this when you have music therapy with depression they stand around the schizophrenia don't know what to do yeah because they cannot synchronize because it's temporally imprecise this is here and this is one of the best replicated findings the decreased phase coherence and we combine that also with computer computer computational modeling to see it's really the phase offsets but this is diagnostic specific you don't see this in bipolar on depression we have a paper in revision and molecular Psychiatry on the next one I know I'm almost over so here now you say how is your face coherence the millisecond related to the more longer time scale of the uh autocorrelation ring so this is uh again now with this here we could replicate there's another data set replicate the autocorrelation window could replicate the decreased in the trial phase coherent now how are they Linked In The Healthy subject the black line they highly correlate yeah if you have shorter autocorrelation window you have higher inter trial phase cohere because you're not tempted if you have a shorter window temperature segregation you're more totally precise interface onset to follow these thoughts and my speech hopefully and in the depression that doesn't exist anymore moving your ACW you tender window provides a template for your synchronization yeah and that is lost and the schizophrenic patients are literally lost and this is millisecond range next one um so then you say where is this coming from these are all task data now showing you whether there's some face inconsistencies in the resting state how do you measure that so Stephanie very clear guy from Austria he developed a new method a very small band pass five to seven Hertz and you basically calculated the projected cycled time point of the cycle durations of the ongoing secretary and then you can you can measure the how the Deviant so the spontaneous face variability and that spent on his face variability is too entropic in the schizophrenic patient irregular in preside milliseconds yeah this is really many seconds yeah so this is really and you see it's too high entropy here applied in central entropy on this spontaneous phase variation next one we do this more and more also on behavioral data second order analysis of the times next one okay so then close to the end how is that

manifest and subjective experience so we developed a space and time experience scale for schizophrenia and we also now we have in the loop in our pipeline this uh time and space experience scale for depression and anxiety because I want to map this also again differential diagnosis and certain items here particular item number seven so this is a schizophrenic group this is the non-schizophrenic bipolar effective group and this is healthy subject you see item seven is almost specific for every patient experience of temporal fragmentation yeah and premium Mission also he has a premonitional typical item made yeah so that's it's also one of the diagnostic layers which can be used and again same for time space experience and depression next one so next one okay I probably skip that one even we did some drawing on those guys and see again temporally fragmented you see Temple fragmentations everywhere for me next one except in my own brain of course um there's no time at all so temporary Precision next one clear I hope it's entertaining I think it's a revolved for stock of mine uh who's now professor in Queens has this very nice uh heading next one and that also goes with some old phenomenology in in schizophrenia minkowsky uh very clever guys psychiatrist he described it as a loss of vital contact with reality now you see I added the dynamic and that's a temple in Precision if you cannot synchronize cannot encode the inputs in millisecond range you're confused and that carried faster the rest of your break next one so for me it's an encoding that is it okay next one um so here I show you that basically I think I hope I made a clear common currency or we missed an end there uh common currency of brain and symptoms we have different lines which you can diagnose next one uh here uh we have subjective experience which I would like to include then you have these different markers which I showed you special measures template measures which I think yield strong support that there might be a sort of what you say diagnosis of syndrome specificity and that really awaits larger clinical trials next one the next one uh I next one I indicated that it is Major implications for therapy particularly in depression I showed you the TMS stuff individual cortex motor cortex and also we have currently an ongoing study in anxiety where we use a breathing protocol and why the reason because anxiety a fast and irregular breathing thoughts are racing like crazy so they have temporal irregularity and that yields Temple uncertainty and if there's uncertainty too much you become afraid so there's a lot of some computational study for temple uncertainty for uncertainty but it's not really associated with a dynamic or Temple uncertainty that's the way I add so it's the next to what they investigate on the cognitive level so while breathing because I'm bruising long if you breathe very slow you have long face Cycles they're good for integrating your different thoughts and your different stimuli and calm

you down and we see dramatic effect so this is a tentative this is 10 subjects you see this is a back anxiety scale and this is a mind-wandering scale spontaneous mind wondering you see a reduction this is after three weeks after six weeks you see a huge reduction in T1 T2 and important in all subjects all of the 10 Single subjects showed a big anxiety reduction that's unheard of in Psychiatry okay it's pilot data it's early but that's again one of the tools besides the DMS for a large stay clinic right and we also individualize the breathing frequency uh with lots of time we let them breathe some different frequency ranges from let's say breathing rates from five to uh to 12 or 14. and see which and then we have various like at scale and which one is the most comfortable breathing rate for them so we individualized and we later want to do this with the EEG and also with a computational modeling next one yeah and I come to the end uh last talk I probably overstepped my 10 minutes uh yeah and if you want to read this is about a lot of psychiatric case histories is the most recent one both books are very short and I think well understandable thank you very much thank you thank you so much you're very inspiring talk um we will have a time for overall discussion so in uh I think that we can start with our next speaker that is cloud there and we will have I have a plenty of questions for you uh but we will have a full time for discussion at the end of the presenters thank you so much hey super good um yes yes representative you were not that lucky I'm sorry you were not that lucky you know um biological Elementary El cambia um presentations estatica a year compartment instruments definition Etc promoters and foreign publication La Masa inflammatory negrita immunological presentation distance disturbances parallel the physical una correlation a positive trabajo boys relations adults Americans reviews Pacific ocean the regulations resistance in English the Lord mindfulness yeah um processes persona cerebral yes mental Chronicle thank you Ontario emotions investigator procedure destiny mindfulness basis anxiety disorders incentives Highlander is modifications automatically Pennsylvania regulations Connecticut resolution representations beneficiais you know foreign activities meeting discussions these foreign aspects from mental health in different mechanisms of you know brain cells that we are really really looking forward to and Mario make a great presentation about digital condition and this challenge that we Metallica so now we have a copy you can grab a coffee there five minutes something and then back here and bringing all your ideas to discuss we will do it in English yes for for having yours on board in the discussion but if you don't speak English you can mentioned okay analysis I don't know opportunity videos again very much everything else important services this evening anymore so I guess we'll do a brief introduction so if you can see across the different presentations we

are talking about a very um old-fashioned distinction what is physical world what is mine or what is you know um causal explanations versus other explanations and this reminds me in the specifically in the in the field of mental health this kind of decision you know the system between those both Fields they have tremendous effects you know in lateral red syndrome it was a condition that was jumping from Psychiatric to neurological disorder so it was a genetic conditions versus was a repressive masturbation by currency psychoanalytic theory they come back again and was confused with Huntington disease there is another disease and this is just because we don't have good models to understand the similarities the same way that yours is you know connecting Brian and means we can do in psychiatric and neurological conditions or in mental health in general we know a lot about from influences exosome external influence lifestyle choices to internal mechanism mechanistic Pathways inflammatory endocrine immune microbiome Etc but also we know that we have tools theoretical tools that can integrate can be dimensionally applied between the both sides and also interventions that can be applied between both sides so it's a it's a great start I have a question for George one in Israeli and so I just want you to bring your voices uh and it's a huge opportunity for make questions for the speaker but also for provide your own insights so I I will open the discussion right now if there are no question I will start with George uh can you talk a little with how your approach can be extended to neurology I mean we have been working together I'm pretty sure you have some ideas just to to bring this discussion about you know psychiatric and neurological conditions yeah first and foremost of course you have depression and anxiety in your patients too from the temporary dementia Alzheimer's dementia minimal cognitive impairment probably maybe even in the subject of cognitive impairment which would be an interesting candidate um so definitely I would screen the literature for that and that's of course should be it's a syndrome um so that would be I think the most direct link and probably many of these symptoms lead you to the doctors maybe even before the memory complaint and the second thing of course then I would need to go into some of the memory complaints they have and for me it would be very interesting the earlier subjective impairment and minimal cognitive impairments and then I would put look for those memories in Dynamic terms so what does this mean I would for instance we have a study now where we analyze working memory and dynamic terms and I would do the same here and maybe particularly in these early stages uh I could imagine the memory function itself is intact but you have some Dynamic changes and I know that the group by Macintosh former Toronto now in Vancouver they did some of this variability and dynamic stuff in early agent and you see changes yeah

you slow down maybe uh that I could imagine so I would look for that and then you can apply the same kind of matters which I have in particular with the cognitive memory Paradigm I just want to mention that Mario has been developing this binding working memory task versus future integration that dividends requires integration of future I mean color and and the form of the stimuli we have a lot of data including easy and fmri so we can apply your spatial temporal so the idea hypothesis we will have longer time window for binding in comparison to Future yeah yeah I would is a classical working memory task with you know studio and set so we have this this long period to calculate the ACW and yeah so show me the exact Temple structure of the Paradigm I look at the temple structure of the Paradigm because cognitive people neglect that that's an intertrial interval interstimulus intervals I'm very keen on this and then I can tell you how to analyze and what I would expect fantastic right any question please first so a really nice part I was wondering whether you have blood or other people have thought about integrating body signals into the general slow foreign yeah good question um you are absolutely right so of course I would argue in our sixth paperless every postulate that and that's the AC autocorrelations and you're strongly involved in input encoding so here really focus on the encoding not the decoding which is an important difference and they'll need to tell you that one and of course the encoding is not only the external input it also the continuous internal input internal input from the body and internal input probably from strong variability changes within the brain itself because they might be treated like an input like spindles in the case of sleep and one and two sleep it's like a strong internally generated input Thalamus cortical variability change and of course I would argue that they uh processed by the end encoded by the same temporal mechanism of external impulse so meaning I know I made a little detour of course I would argue and we already discussed this in one study link fund particularly the autocorrelation window to the uh heartbeat evoke potential and I know that there is one study for instance for scale free activity it's a parallel exponent and of course you see responsibility to the Apple so it's clear because it's a scale free system yeah and of course the brain will react and that's for me con so if you want to put it in a larger perspective with complexity magic Bruce West a comment on that and connecting with allostatic over lower now after hearing you several times but after this visit I understand the relevance of encoding in your approach right so the the inputs this is so because it's a it's not the typical way that the people interpret the inputs and when we think in allostasis you know it's supposedly the responses to environmental demands but those responses depend your internal organs your Dynamics your internal Dynamics your encoding of this external information so I see better

links as our paper are really submitted to connect allostasis with this internal time scales like you know the internal structure not only to respond to environmental environmental demand but to explain the term anything you know because same stressors can trigger different internal States or even exposure you know you can understand the explosion like external trigger but you have a lot of internal signals you know that accessor beta explosion so if you want to I mean imagine you have a lot of money and you can you can do you can measure wherever you want what should you pick up choose for for an allostatic interest analostatic index sorry is an easy good question is the expectation question the the use of biomarkers in assignment I mean if I have a lot of money a lot of funding I'm going to use several I mean most of the most of them the the the the primary ones the I mean all of them because all of them are related that I can't find we can find them the distance type of uh allostatic index wait wait because maybe I was I was saying too much imagine you don't have all the money so you have to pick up three of them what should be the best [Laughter] yeah I know how to what can I use um the first one is the second one something how to say cheaper this is a how to say wait wait yeah and the third one is the moon inflammatory proteins directive protein thank you so much for your questions please please because we have guests online okay uh when I when I when I started your presentation let's talk about this speech uh uh brain modulation and people with depression I I thought well this might probably be change so how about then you show that you have used TMS uh to kind of restore uh or try to so then and I was wondering if you have thought or whether it would be feasible to use biofeedback some form of biofeedback uh in this in this patient with depression to kind of speed up like the entrainment maybe or what are you thinking about well I think in EEG based by a new configuration by your paper because you show the frequency changes as a byproduct of depression and that's why they tend to process things lonely and so on and so forth so then if we feel this information bad biologically and there are many ways to do that now could it be possible to kind of help them bring them back to track in terms of speech at the neural level and therefore hopefully at the clinical behavioral level as well yeah I mean that's exactly the idea so in uh that's what I would do if I would individualize so we individualize the region uh now I'm really looking forward to a large clinical study and there I need what you just promised a lot of money or facilities um and I cannot do this academically um that you really have say okay you have uh three arms study you have either motor cortex visual cortex or dozolateral prefrontal cortex according to the neuronal and psychological deficit so you have a working memory task following the dialogue letter you have a motor cortex task which I didn't show you a motor tapping

Precision error which we can do uh this is established and we have the psychological visual tasks as I showed and then the next thing you want to do EG and fmri and Method Dynamics into a power spectrum of those respected regions okay and compare that also with the behavioral data so you do a Power spectrum of the behavioral data two for instance of the finger tapping how much they're lacking behind and then you have basically the power Spectrum on the behavior level as well as on the EEG and on the fmri interest loan range and then you get sort of an overview of different time scales and see how much are they relate with each other you can do Mutual information you can do a vector Auto regressive uh model for linking these different time series and then you see how they are linked to each other so that gives you sort of a fingerprint of the neurodynamic basis of their behavior and I'm insisting here on the behavior because so much in therapy goes either pure on the psychology or pure on the neuronal and that's Nick and I would like to even include my subjective experience skills I'm after all a phenomenologist as a philosopher where I'm coming so also this depression which I didn't show you the time space experience scale of depression so if I have that and then I have the data to go into computational model so then ideally you clever guys you converge my EG data my individual EEG with the computational remember that the two red points dots we were discussing yeah so then you can simulate how I can change that particular neuronal Dynamics PSD or whatever or the correlation Windows scale and of course you probably also want to have a dynamic autocorrelation window which I didn't show here which you need to have and then how I can change it how I need to tune the computational system to change that frequency and that way we again for each subject differently which I suppose depends also on the dynamic repertoires that's the variability which that's why I wanted Dynamic acid and I think that's bringing the biofiber to a modern level and then he can give me the exact data and then I come to you and I don't know you seem to be a social scientist but maybe you can build me the right machine where I can Implement that within continuous online feedback system so I start with the frequency which he recommends me the Red Dot how we can shift it yeah and then we can build this yeah but I would not do this before because now the current data on BCI and TMS are at best individuals towards personalization because you really have the holistic approach the AGP you're not just a biological the psychological or the environment so what I can do this could be run in a larger clinical studies either the TMS this is 3M TMS end with the computation model or for the breezing the breezing might be a slightly easier thing uh the only thing in the model then is how do you apply a breathing pile to the model that's not an easy one because the input is the

real problem of the formalization of the input on your side yeah we discuss this in our group we have no solution to that oh because you just you do yeah yeah so that's sort of no I think we need to make a discussion between breath and mindfulness but first Carol have a question um so can you hear me no yes okay yeah um so I have two questions for two speakers so I will start with York um so you started your your um discussion your presentation saying that models uh addressing genetics and disease are not enough so um so you said in your presentation that models uh use in genetics and disease those kind of models are not enough I'm wondering if we are missing some parts so um since my I have a background in epigenetics I'm wondering uh well these epigenetic changes have risen as a mechanism connecting um genetics of a person and experiences environmental stimuli so I'm wondering whether you think that maybe genetic regulation through epigenetics can add something to the topic it was against the case number okay the case numbers are not substitute and I used falsely now genetics as an example okay anyway so first of foremost I'm not an expert in genetic epigenetics I cannot say much so this is so everything I say it just comes from a very bird's eye view from somebody who has no expertise in them um but just say coming from my clinical side now I speak as a psychiatrist what is clear life events play a key role in that in different ways first they triggers basically all depressive patients therefore and it's not the objective language it's a subjectively perception of that life two different two Claudia and I two different people same life events exactly the same life even I become depressed she remains no okay so don't tell me why that is so and that's a problem because that depends and that's why I would argue remember this one slide where I had the childhood trauma in the time perception that's the thing yeah so and that I think and then that's a trigger in that but then you have also more concentrative life uh trauma continues trauma you become depressed or you react with the PTSD or post-traumatic stresses or whatever and of course epigenetic changes like absolutely clear though that's what so far I can go I cannot go further because I'm lacking the expertise I can compliment a little bit on that up sorry so first of all um I think that we have a bias with genetics we tend to immediately forget that the genes need expression and genes are inside an organisms you know this is so important and what you start to thinking about in more complex genetics right like cims or for PRS polygenetics score right the interaction with environment is critical and when we talk about epigenetics oh my God you should be interested in that because one important thing is epigenetic looks is Dynamics is is again temporal Dynamics and how the burden across all the lifespan modulate this clock interact interacting with the genetic and depressed position so as you know we need

more of this kind of dialogue you know because um in red light we do a lot of genetics and it's incredible they huge opportunities we miss to connect the dots from environmental factors to genetics and then to this important aspect of subjective experience right we need to change that dialogue if you can provide time scales with enough data points yeah I mean I'm talking at least 100 data points if not more then we can do the same measures yeah yeah like autocorrelation we know one of my favorite as you probably figured but also other sample entropy what we did in this case exactly as you could do because after all the different levels must communicate with each other because my assumption is that's tempting so it's the encoding process between brain environment is Temple then I'm I'm it's a reasonable assumption that maybe the encoding process from the inner of the organism and the different layers is also done yeah I make a very strong claim imagine but then meaning we could link and the ultimate is that should be scale free organization so the same kind of pattern which you see in your dynamics of your epigenetics I should somewhat see similar pattern in my smaller time scales of my brain time so that would be great exactly yeah that's okay free so you provide a large Russian doll I just have a small one compared to New York Times maybe but I don't know but this can bring you know special temporal Dynamics beyond the brain because yeah temporal metrics of different organs modulated by the epigenetic Rocks but it's a temporal code exactly connected it's a temporary coding process that's why I'm kidding yeah English is our common code now it's the only one I don't speak Spanish yeah uh so there we are yeah if you don't that's like the temple quote in the brain for me with the brain body environment so the principal difference I'm sure that the brain you know I always think brain sits here and listens to us and says how stupid are you to separate me from the body and the environment it's just one thing for me it's just a matter of balances it's continual and that's what's a patients yeah second question yeah I have another question poor Constance at this time um so I was seeing your presentation with great interest and you mentioned some different effects of a mindfulness on different biomarkers current metabolic by markers stressed by markers Etc yes um so I'm wondering whether there's maybe some study you have read maybe you you are familiar with since you have a background in Biochemistry um have you seen any study reporting any gene expression changes uh related to mindfulness Okay so I don't know if I think okay great thanks the exactly mechanism how this happened I mean the the most close we are is that we have some pattern of activity in the brain that for instance you encode for something that is kind of very real or not real and there is done in going there and that information uh goes down through the hippo AKA totalom hypothesis axis and you

generate uh um you know a solution of different hormones and that's it so that is kind of how the things happen and then you have the cortisol that Claudia explained to adrenal how this happened with dms is not very clear there are um genetic epigenetics modulations but not I mean is to the level of gene expression but not Gene like modification of course and uh and this is related to the measurements is more that have been targeted is in Pro cytokine inflammatory expression and interlockings uh also and um yeah kind of this range of like pro-inflammatory environments in a way and this is true uh yeah mutilations and and this is the mechanisms methylation so yes but exactly how that is orchestrated uh is is not really very clear um because also there is some confoundings in I must say and disclaim in meditation studies especially with the advanced meditator studies where a lot of these findings that has to do with the lifestyles that you begin to have when you engage in meditative or complaint contemplative practice like um the degree of awareness of your elementation and like bodily awareness making sport or not that that also generate like an holistic change through pro-inflammatory I mean decreasing this pro-inflammatory environment or stressful Lifestyles that we could have so there are some um short-term studies with eight weeks showing already epigenetic change and also in telomerase is the minimum time for expression but normally the the biggest effect is for more for longer time and then you begin to have a confounding with lifestyles in general um so yeah one more little thing uh I'm curious for myself uh is there any like type of person that is more prone to meditate like easily to be like Advanced meditators is there any praise position or some some time or some type of person that meditation comes easily for them that I really I I'm not sure of I mean there are some studies about relating the week five with like because you have like some like great mindful um outcomes you know you have like State mindful or trade there are people that is normal more mindful like if you may if you pass to people in this room the mindful scale mindfulness scales there are more people even though they don't meditate that they're going to have a more uh like mindful trait yeah but people that normally approach to to to to to meditate is people that is normally more anxious and have normally more prompt to be anxious or to be stressed and people look for meditation when they are in uh not very good mental health shape which is natural it's like looking for medicine but in a way of practice in a way so I would say that people that approach to meditate for mental health reason normally is a select group of anxious people and but then this has his his consequences of course and um but there is there is people that is more disattached or that have less cognitive Fusion that is more like mindful as a trait in a Continuum but I don't know if for these people it's more easy to meditate because always there

is some I mean there are obstacles for each type of people right so there are people that it's going to have some fights with their like solidification of thoughts and there are people that have a profile that is more evitative so it's his evitative patterns of thinking are the the obstacles for them when they come from their meditation so so as you may know this question that Carol has made is very important because probably you are very aware about this I see this nature communication or nature of human behavior paper showing that the effect of mindfulness are not such as strong as we believe and there are a lot of failure so but the question is personalized medicine so what's the degree of of mindfulness or breathing or you know if we are thinking about mechanisms for self-regulation perhaps that is one of the mechanism associated with mindfulness or producing allostasis right why we don't think in multi-component interventions in in the field of brain health the multi-component interventions are the best no comparison with any other across all all possible interventions so maybe exercise breathing and mindfulness combining and personalized right so maybe you want to mention your how you you know personalize the assessment of of breathing so we can but first let's start with your opinion about this yes sure so we we publish a paper last year actually how meaningfulness-based intervention were difficult different brilliantly differently and uh efficient for different type of mindfulness based Baseline people so um for sure there is there is a thing of uh baselines where people arrive with uh and also there is a timing that is very important for instance for the things that I show for depression it's not when you have the person in the more acute episode of depression but is when they are really for relapsing and so I think is very important what you mentioned because actually what you are fostering is a kind of a mental culture of um not going to the doctor in a way when you are like completely done and when they okay great sure um and so so you are fostering kind of a more preventive a moralistic approach so you go you can turn whatever you need to do more exercise or or have a better element station or if you are like not sleeping too much because you need to do this you need to then you kind of counterbalance meditating or whatever so yeah hopefully we arrive to something like that but to prevent like Dementia or all these things normally you have to to try to sustain the better lifestyle uh you could do whatever so maybe it's useful to do some kind of it will be accident but to look kind of a lifestyle index com like that you could okay that you can kind of put there your credits on meditation and running and stuff and and to see like basically how much drinks you're gonna have in the weekend without like you know compensating your oxidative kind of yeah interesting comments I mean first off sir I think one thing which is really neglected in the meditation literature is the induction method and the data

on the neuronal side extremely heterogeneous because you have all kinds of induction methods I was in India last year invited um I mean you can't imagine what kind of breathing techniques they have through the nostrils different nostrils and it is insane and of course when you're there you have to undergo all of that because it's only experience counts lots of knowledge it's incredible so anyway so I think that's a key thing so different people prefer different induction methods um that's of course and I would argue the meditative state is a common final function of the password I know that some review of our papers killed that figure we had a nice figure different induction they all lead to non-duality and I would keep the induction techniques in a very wide sense for me for instance I'm a runner so I like run and women in nature that's like a meditative state from a flow statement so I would Define it very well so meaning that's where you need to individualize and now we actually investigated where one paper in preparation where we really look for the different kinds of induction techniques how they because that's input encoding it's an internal input which you encode the breathing input oil mental input how that relates of course to my beloved measure of ACW and one of the reasons why I'm so focused on this ACW is also pragmatic reason we have other measures parallel exponents in they don't yield in those context they don't yield those results for other yes you would see this in order so that's internal input encoding that leads back to the question which you raised for the heartbeat and the intrinsic neural time scale yeah so also the internal input is in code so now I would then basically let's say again I would do a time scale profile maybe a power Spectrum then of course I need to better provide better neural phenomenal connections but then I would go from there and individual and that maybe that induction method is the right thing for you yeah so that's the way I would individualize and one thing sorry to Reign into your parade there's a lot of aversive effects in meditation so I'm as a clinical psychiatrist who wants to apply meditation to psychotic patient needs to be very much aware of that yeah and again last thing so that's again it's very important who gets those adverse different so meditation is not just good can also be bad like everything like the pseudocybin can be very good it can be a disaster well you have the wrong context yeah so in last comment on that so when you're in India the people smile on the because they say that's not meditation and meditation is not Buddhist and you also said mindfulness meditation sort of I think you use the terms interchangeably it's mainly for let's say people with anxiety of depression that's the Western world meditation has a very different context when you go to China or India completely different yeah so very very important and what is presenters in mind for okay now I do five-minute mindfulness and then I'm uh able and then I'm more

capacitive and can have more success that's not the context for mindfulness so I'm really I live over there so I know how they do it I'm strongly against it it doesn't work yeah and you can see this is let us through that context about the challenging effect of meditation they are recognized as any prescription or whatever and has to do as you say with the with the person and um who is prescribing the thing so normally panic attacks and and and yeah difficult events that happen during meditation has to do with is very similar to exercise I mean you normally if you decide to run a marathon you are not from nothing going and go to Rana Marathon because probably you're going to have some you know some um problems with your muscles or your knees or whatever because you are not prepared so normally you have to prepare prepare it and go very like increase the frequency very slow and everything and normally panics attacks and all this kind of secondary or what is called The Dark Knight of the assault or meditation has to do with these doses that is people from the West that goes and from being really like running ahead with their jobs and everything they sit and they have this panic attack when they sit for two weeks or something like that and also when they are in a very bad depressive or Society State and they sit to meditate then they have this thing so this secondary effects happen basically for a bad management and prescription of the technique but it's not the technique itself and regarding what you said about the the how mindfulness is seen in the west I say that actually that mindfulness was integrated in there was to make frontal stress and Lifestyles from the westerners uh yeah point of view you know like anxiety depression as kind of a prescription and in Oriental countries it's a lifestyle to live like that and and and that's why I'm saying that how we what we are trying to achieve is not this only mindfulness stuff but that's why I'm um evoking for something more holistic that has to do with lifestyles in general where meditation is one component that helps you to be aware of your habits and helps you to change your habits uh because you become better to sense yourself and and since the rest of the people basically yeah thank you so Rodrigo is not paying attention to the computer and is absolutely insightful about the discussions so do you want to make any questions now that you have done a review and kind of characterize the the status quo the same so I I guess you're getting ready to kind of start using this approach and they are wondering about uh red light whether because freelance has been Gathering a lot of data and I get from the existing data will be possible to draw some of these Arrow static measure so then are you planning to do that and so remind me red light adventure started before the Copic pandemic or you had started after we started before because the meaning that you have before and after yeah but no because for before we have only pre-existing data okay that was

collected previously I was wondering because just wondering would be great yes well in any case at least you could look at the because you have you have environments thank you Marie yes uh we are planning to do a lot of things with the plasma sample the first tool um we would like to to evaluate the the the most important or more classic and neurodegenerative biomarkers and of course if we but also we want to evaluate this type of holostatic biomarkers well if we have money of course we are going to evaluate everything but I think that now we have some preliminary data from the the the collaboration because the collaboration with the Andres Ietpsi and we have this type of uh immuno inflammatory proteins but yes some of some of them are related to this allostatic low biomarker but I mean we can use it for the renew one and everything and is here in the in this web preliminary data we have a healthy Contours the MCI or the Euro Golf Course patient and also ID I don't know if we have some FTD patients but probably and now we are planning to do something with um I'm sorry and we are planning to do something with the car pounds and with an association with the NIH and uh yes we are going to have some preliminary attack for the renewal of the R1 or the red and I want to add two things of that we are also starting to do exposome mediation like social epigenomics effects in the methylation and and many different epigenetic blocks that by the way many of those doesn't work in our population you know it's a very important and really reported Hispanic Paradox is scholar so the patients are younger some clothes related to aging I working well and showing the strong effects of the environmental factors accentuating the chronological game gap sorry this is one and the other thing is we want to do allostatic intercepting Network in all our patients we are going to do that and we we we just started to I would say change the idea of predictive coding in a linear way but using his special temporal approach and I want to try it in their static interceptive Network for for the frontal temporal dementia patients of course yeah yeah we are going to do it yes so now I think it's fantastic value it's working it does work but I wonder if you have about this idea of what you all represented about always happy loads and mindfulness because I say once I think so it doesn't work here of mine also some people is also work for other but can we go through the subjective experience of mindfulness to some objective metrics that can allow you to identify who is objectively benefiting really from this and follow that all because that's that's not really is very specific approach is it what do you think yeah I think would be ideal to be able to do something like that but I'm not sure yes yeah yeah um yeah I I think uh you can have I mean you can have of course you have an interest subject variation that is like killing the biomarkers like uh findings and everything but you have people that is more successfully regulating and

people that will take more time and normally you do the studies and you say Okay um these people in eight weeks they didn't manage or they manage or they have different stressors from the outside so basically what would you try would like to do is to do an internal Mark of one that people is managing their self-regulation and everything and so you can use each one as their own start points and and end points and if some people need eight weeks good for them because all programs and in in the US like mindfulness-based cognitive therapy is covered for health insurance and stuff but people needs more time and people who is under more allostatic load is going to need a more maybe a steep uh or not so steep meditation apprentice apprentice and other people so basically to make it short I think will be like uh some internal marker uh when people is basically regulating and he's in place with himself and that could be subjective but will be also nice to have like a more like biological marker because people also lie to themselves so to the more biologically driven uh approach to efficacy of this particular intervention yeah but the subjective part is also very important because there is people that can be with a lot of cortisol and they are like yeah they are doing okay and then they they just burn out when one day to the other so how do you say I mean subjected the ultimate goal in the community in terms of how we can really corroborate biologically and if that happens and everybody equally vulnerable to implement biological every influences on mindfulness and if they are can they carry forward what is beneficial attack on long press so that this demolish is lackings biological to bring the community together and solve this lower standing yeah yeah yeah so yeah sure maybe to start to do some kind of clustering profile of you know kind of people with uh subjective and like more lifestyle stuff and also meditation so you can do kind of a prediction within a more like small like yeah group yeah yeah sure yeah yeah other questions I think the computational core are very far away yeah be centered hi first of all thank you very much to the speakers uh great talks I was going to pick up this last discussion uh I think that I really agree with with Mario that there's some like biological mechanisms lacking the relation between the environmental burden over neurodegenerative diseases and mental disorders we can have for example the case of the depression it's higher in in higher socioeconomic status countries as compared to lower socioeconomic status countries and the same for neurogenitive diseases right the inverse sorry reimbursement so we have some like peripheral biomarkers that we can use to track these diseases we have some like Central mechanisms that are related to these diseases and these mental disorders but there there's a lack of the link between the periphery the environment the exposome right and the the internal Central mechanisms in the in the nervous system so I think for example it's a it's a good

example to use the the case of uh of meditation because it's a very uh like specific effect in metabolism for example we have an increase modulation of oxygen metabolism in the brain is it maybe related to I don't know like blood brain barrier permeability for example there is facilitating the the entrance of oxygen for example and maybe that could be related I know with apo-e Gene that is Rachel cardiovascular risk and permeability as well so all of those questions like have to be like unified in a common framework I believe this is not just a general commentary not a specific question sure I mean that is the you know this cognitive I use cognitive um yes I mean this is basically this this is this intra individual you know like uh cognitive protective factors you know it's like it's the history of the system that is giving you also that variability and is the one we can basically we cannot explain I mean we want to we want to make it the most predictable possible so we can take people on that track but at the end you don't know we are still modeling this into individual difference that make people more resistant or not according to their lives in general and I think that your point is is an overall question so it's happened right now and we lack theories we lack evidence you know and it's not only to bring cow cell explanation it's connecting the leverage because this is the main Challenge and it's an academic barrier it's not a it I mean in nature this happened together but we have been used to study you know in a very isolated way I think that we as we are starting to do it and I see that theory I I know that yours is not a fan of the philosophers but we need to bring theories that can integrate levels neuroscientists cannot do that at all home they are not prepared for that on the contrary they are prepared you know for the micro level and isolation you know internal control isolation that's it well we need here is that really really integrate the dots we are just you know doing a small dots of that but we don't have a common approach for that because of that I like anastasis by the way but yeah so don't misunderstand me I'm not against philosophers I'm against particular football yeah yeah I know I know I'm currently dominating in the Un-American match for the meditation so that's um we develop what we are going back to that example so we're going back we indeed develop a time and space and self-experience scale yeah so for meditation same what we do in schizophrenia yeah because that's a phenomenological correlate but again I think probably what you really want and let's say for the individualization when I have a tendency perhaps depression and are very slow probably a slow induction technique is not the right thing so that's where could be potential markets yeah and you can try titrate The Experience yeah doing the different induction techniques what is the best yeah and that will be an individual fingerprinting of the induction techniques I think that's absolutely key um and then with that coming there from the

phenomenology that will definitely give me also some idea about neural markets of course I I'm not sure but I don't think that sort of what we do with the global signal topography what I expect of course is clear I like expect less topographic differentiation in the global signal meaning all regions are more homogeneously characterized by their Global activity that's why you don't process the inputs in a more differentiated way because the brain is not literally more synchronized so that could be potential markers if you can link them to your degree of let's say Duality non-duality um yeah the degree of environment self-distinction how much you make a this is a great this is tension you know environmental self is you know is across so many other paths we need to to connect I think this is this is a very missing area well again you have any questions for my Kim yes well it's everything very interesting I was thinking about the relationship between meditation and allostasis specifically by the modulation of the insular activity this is a region the main region of the allostatic intercepting Network so maybe this can be a hub of influence from to meditate and therefore regulate the activity of the insulin therefore regulate the activity of studying perceptive Network therefore regulate the organs and stabilize the organism could be this could this be possible how is the evidence Associated meditation activity of the insulin because that is a question because I reviewed that evidence and only two papers one Frontiers made that Association but is that theoretically um scientifically based meditational insular regulation there is a lot of discussions and normally you will have to you will have something that support meditators have more interceptive awareness connectivity but um there are there is another yeah so the the short answer is yes but there is still lack of some replicability because of the degree of the training so normally you have these studies in very Advanced meditators that are not they're not the real Lifestyles that we can achieve living in in in in a city basically and with normal jobs and stuff so but normally yeah you take on the best month and if that is the case then yes and and there is a sense of self-sensing and self-regulation in all the models on the um there is a self-self awareness and self-regulation that is increased and insular participate in this same thing part of the awareness so I think and and how to relate to allostasis I think that it's something that we have been all thinking I mean one stressor will hit you in a very different way and how your internal system is standing right so and how your internal system is standing that is you can train to regulate to keep some kind of consistency in how to maintain your like mental stability or emotional stability you can train for that so when you um train enough then the stressor will hit you in a kind of a good shape that you can absorb this and it's not going to create some critical point that is going to to um to trigger some other you know Cascade of inflammatory

and like that so you can kind of prepare yourself for allostatic load um but yeah so that is what I will yeah because I have like a two-step question I'm gonna make the first one I'm doing the second one we got the same one it's for you okay about the is mindfulness how do you considered the Micromax effect against for instance like hyper pockets it's safe asking for a friend in attention or disorder for reasons for you to be like an inner statement no no I mean Focus attention you can use an internal or you can use an external object and doesn't matter the thing is that you the exercise is to keep your focus on the object of choosing and realize that you distract and come back the the attention to that and doesn't matter if it's external or internal you can train both of them but there is also some papers coming from Feldman and a guy that is very like it has a great literature about how the processing of respiratory Rhythm and everything and he's doing a uh animal model of meditator mice that only regulating the you know the Rhythm through some drugs manipulating only the the breathing system he's making the the ratless anxious and and perceiving all kind of things so I would say sure I mean literature shows and also in meditative Styles you have a external focus and internal focus and in terms of like cognitive exercise of coming back your attention to the focus will be the same but of course that there is a link with the breathing that comes from a more bottom-up point of view uh that is not that we haven't like separated yet to establish if there is some different efficacy in the training with breathing or with the or with the external object and the second one is a short question for all the speakers and if you have issues like a keyword like a key idea that someone that doesn't know anything like me for instance like a a you know dealer you can take and make a change in their lifestyle then most important like a short no I don't think that there is something because it's something that we have been discussing a lot of the personalization and you know one solution is not is working for an individual that is not working for another like mindfulness can be very disrupted for some participant it can be very helpful I mean very healthy healing a lot so I I wouldn't bet for for nothing uh in particular I would say of course we had you have the least you can repeated that but this is a question we tend to think this releases reversal and can be applied to everybody and I think that we are going to show enough for coming very important paper the risk factors interacts in a very calorie in the paper interacts in a very differential different weight in underrepresented population and the combination and factors for preventing or for causing or for influencing the risk it change everything so yeah but yeah this is a great question I think one area called Behavioral Sciences I'm focusing on that particular area because everybody shows that there is a capital awareness knowing what is good for you and doing so really committed to do that so

the behavioral changes all developing program that women should people who all right and the wedding to developed embark on actual Community personalized behavioral change towards Hair Design so you see that for example this is because we talk about many of the decisions we discussed here today prevention and there is no prevention we don't shift people understanding from this type of Behavioral changes and every lifestyle factor that could be both bad but it's so disagree and from there to the core prescriptive type of mind that we have in our country where we are very much focused on children our Premiership America even management many by the commercial mention of our medical system but I think there is a lot of work to those and Learners about that particular aspect that you're talking about because it doesn't really matter how many solutions we develop how many tools we provide the people the people and this one of my colleagues actually the director of the schedule the Center for Behavioral changes in Harry Potter University of the Scottish population actually about how where people were who what drivers of brain health uh including all these things so then I'll ask them a lot of questions about that and so the majority of the population were aware of these changes so but when we ask about are you ready to go for it people say these people say yes and when they have one we say well if I get sick because you see that the video is not really prevention it's just pure I mean nothing people of mind the reasons we don't shift people knew about what is really prevention not only people do our health systems views and approaches to that so we are running out of the time last word Claudia and yours use the mic if I have to choose some word be happy and that's it well I am done I don't know anything to them so thank you so much our guests are going to be here in case you want to interact with them to make questions for today right we are both here so we welcome you to to interact with them they are making collaborations across the globe so enjoy them thank you so let's bring to take a picture with your mind